Week 1 Workshop Handout
Gradients, Curvature, and Potential-Field Intuition

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Session plan (first 4 hours)

1. (0–15 min) **Course introduction**
   What the course covers, weekly rhythm, assessment structure, expectations, office hours, and where to find materials. Download mateials while instructor goes over outline of course.
2. (15–45 min) **Back-of-the-envelope (BOTE) calculations**
   In groups of 3–4, estimate momentum $p = mv$ and kinetic energy $K = \frac{1}{2}mv^2$ for:
   - **a.** a tectonic plate
   - **b.** a freight train

   To estimate these parameters, you will need to know/learn a few things about the structure of Earth, the physical properties of common rock types within each layer, and make assumptions (understand ranges) for required values. You will need to keep track of units and become comfortable with uncertainty and order-of-magnitude estimation.
3. (45–60 min) **Q&A from online lecture (15 min)**
   Address questions from lecture; otherwise give a concise teaser: scalar vs vector fields, ∇h as direction of steepest increase, and ∇^2 as curvature, vector field = gradient of scalar potential ($F = \nabla\phi$).
4. (60–70 min) **Sheet demonstration of potential fields (10 min)**
   Edge holders impose a boundary height (Dirichlet BC). Pushers underneath mimic sources/sinks (Poisson region).
   Prompts:
   - Where is $|\nabla\Phi|$ largest?
   - Where is $\nabla^2\Phi \neq 0$ vs ≈ 0?
   - If we "move our observation plane up", why does the surface look smoother?
5. (70–75 min) **Micro-break.**
6. (75–105 min) **Gradient Workshop (paper, 10 points).**
   Please bring your printed Mount Shasta / Marble Canyon sheets. Work in groups of 3–4.
7. (105–120) **Debrief.**
   Discuss the answers.
8. (120–130 min) **Transition to practical**. What you will start now and finish in lab/home; how to submit.
9. (130–230 min) **Practical start: Topography notebook (programming, 10 points).**
   Everyone loads the ETOPO NetCDF, plots a regional map, *writes* an explicit 5-point Laplacian, and produces a $\nabla^2 h$ map (smoothing/profile optional starters).
10. (230–240 min) **Debrief**. Share one finding where $|\nabla^2 h|$ is large and why it would blur first. Reinforce: farther ⇒ smoother; closer ⇒ sharper.
11. (230–240 min, time-permitting) **Preview next topic**. Geophysical fudge factors... i.e., physical properties of Earth materials.

What to submit (Week 1 practical; total 20 points)

A. Gradient worksheet (10 pts). Submit a scan/photo/PDF of your completed sheet (arrows, notes, short answers). This will be done as a group, but submitted individually. You may all work off the same worksheet.

B. Programming (10 pts). Submit your notebook or `.py` with *your own* 5-point Laplacian, plus: a regional map, a $\nabla^2 h$ map with symmetric limits, and a 3–5 sentence interpretation (why $|\nabla^2 h|$ highlights sharp features and why increased distance smooths).

Rubrics (concise)

Gradient worksheet (10): correctness of arrows (4); thoughfullness of written answers (4); accuracy of written answers (2)
Programming (10): code correctness (2); figures (2: clear maps, symmetric limits, labelled axes/colorbar); interpretation (6: thorough and thoughtfull answers to questions).

Tips

- Work in mixed-background groups; geologist + physicist + ...
- Share your expertise freely and use each other's knowledge to fill gaps in your own.
- Work colletively and constructively towards the common aims. Value each other's opinions, feel free to respectively discuss differences of ideas.
- Keep your regional window modest for the computer exercises so computations render quickly.